

CCNA 200-301 Notes

Chapter 33 — VLAN Types & Switch Management

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Based on the supplied Chapter 33 video summary and command list, organized into concepts, configurations, command purposes, verification, exam notes, and troubleshooting.

1. VLAN Types

Type	Purpose
Data VLAN	User-created VLAN for normal user/data traffic, such as HR, Sales, or IT.
Default VLAN	VLAN 1, pre-configured on the switch; ports belong to it by default. The video states it cannot
Management VLAN	A VLAN used to provide remote management access to the switch, such as Telnet or SSH, ins
Native VLAN	The VLAN used for untagged traffic on a trunk link.

2. Data VLAN

Used for normal user/data traffic. The video examples are HR, Sales, and IT.

```
vlan 2
name HR

vlan 3
name Sales

vlan 4
name IT
```

3. Default VLAN

VLAN 1 is the default VLAN. Switch ports initially belong to it. The supplied video states that the Default VLAN cannot be deleted or renamed.

4. Management VLAN

The video uses VLAN 4 (IT) as the Management VLAN, assigns an IP address to interface VLAN 4 (the SVI), and then configures VTY lines for Telnet.

Concept: Management VLAN → SVI → Management IP → VTY → Telnet/remote management.

5. Native VLAN

Native VLAN carries untagged traffic on a trunk. The video emphasizes that Native VLAN settings must be handled carefully and consistently.

6. Creating VLANs & Assigning Access Ports

Create/name VLANs

```
vlan 2
name HR
vlan 3
name Sales
vlan 4
name IT
```

HR ports

```
interface range fastethernet 0/1 - 2
switchport mode access
switchport access vlan 2
```

Sales ports

```
interface range fastethernet 0/3 - 4
switchport mode access
switchport access vlan 3
```

IT ports

```
interface range fastethernet 0/5 - 6
switchport mode access
switchport access vlan 4
```

7. Switchport Modes

Mode	Use
Access	End-user device/PC connection; assigned to one VLAN.
Trunk	Inter-switch connection; carries multiple VLANs using VLAN tagging.
Dynamic	The video describes automatic negotiation of access/trunk behavior based on the connected device.

8. Trunking & Tagging

802.1Q is the standard VLAN tagging method covered in the video. The tag identifies the VLAN while the frame crosses the trunk. ISL is mentioned as an older Cisco-proprietary trunking technology.

```
interface fastethernet 0/24
switchport mode trunk
```

9. Restricting VLANs on a Trunk

The allowed-VLAN command limits which VLANs are permitted to cross the trunk.

```
interface fastethernet 0/24
switchport mode trunk
switchport trunk allowed vlan 2,3,4
```

Purpose: allow only VLANs 2, 3, and 4 over this trunk in the video example.

10. Management VLAN + Telnet

Complete configuration shown in the video:

```
enable
configure terminal
interface vlan 4
ip address 192.168.1.100 255.255.255.0
no shutdown
line vty 0 4
password cisco
login
```

Command	Purpose
interface vlan 4	Enter the SVI for VLAN 4.
ip address 192.168.1.100 255.255.255.0	Assign the management IP to the SVI.
no shutdown	Administratively enable the SVI.
line vty 0 4	Enter the VTY lines used for remote access.
password cisco	Set the VTY line password in the demonstrated configuration.
login	Require the configured VTY line password.

11. Important Concept: Management VLAN ≠ Authorization

Using VLAN 4 as the Management VLAN provides a management interface reachable through that VLAN. It does not automatically mean that only the IT department is authorized to Telnet to the switch.

Restricting which source addresses/users may access VTY lines requires additional access-control/authentication configuration. Those commands were not part of the supplied Chapter 33 command list, so they are not added here.

12. Verification Commands

Command	Purpose
show vlan brief	View VLANs and port assignments.
show ip interface brief	Check SVI/IP addresses and interface status.
show interfaces trunk	Check trunk status, tagging/encapsulation information, and allowed VLANs.

13. Command Cheat Sheet — By Function

VLAN Creation

```
vlan 2
name HR
vlan 3
name Sales
vlan 4
name IT
```

Access Ports

```
interface range fastethernet 0/1 - 2
switchport mode access
switchport access vlan 2
interface range fastethernet 0/3 - 4
switchport access vlan 3
interface range fastethernet 0/5 - 6
switchport access vlan 4
```

Management VLAN

```
interface vlan 4
ip address 192.168.1.100 255.255.255.0
no shutdown
```

Telnet / VTY

```
line vty 0 4
password cisco
login
```

Trunking

```
interface fastethernet 0/24
switchport mode trunk
switchport trunk allowed vlan 2,3,4
```

Verification

```
show vlan brief
show ip interface brief
show interfaces trunk
```

14. Exam Notes ☐

- Data VLAN = normal user/data traffic.
- Default VLAN = VLAN 1; ports belong to it by default.
- Management VLAN = remote switch management.
- Native VLAN = untagged traffic on a trunk.
- Access = end device / one assigned VLAN.
- Trunk = inter-switch / multiple VLANs.
- 802.1Q = VLAN tagging standard covered in the video.
- ISL = older Cisco-proprietary trunking technology mentioned in the video.
- Allowed VLAN = limits VLANs crossing a trunk.
- Management VLAN by itself does not restrict which users are authorized to Telnet.

15. Quick Revision

VLAN Types → Data / Default / Management / Native

Management → SVI + IP → VTY → Telnet

Access → End Device

Trunk → Switch-to-Switch → Multiple VLANs → 802.1Q

Allowed VLAN → controls VLANs crossing the trunk

Verify → show vlan brief / show ip interface brief / show interfaces trunk

16. Troubleshooting Checklist

- show vlan brief → verify VLANs and access-port assignments.
- show ip interface brief → verify the Management SVI and IP/status.
- show interfaces trunk → verify trunk operation and allowed VLANs.
- Confirm the required VLANs are included in the allowed-VLAN list.
- Remember that the Management VLAN is the management path, not an authorization mechanism by itself.

Scope Note

The chapter content and commands are grounded in the supplied video summary. The Management VLAN vs. authorization clarification is included to prevent a common conceptual mix-up; no extra access-control command set has been added.